

① Binomial RV

$$X \triangleq \text{Bin}(n, p)$$

$$P_N(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E(X) = np$$

$$E(X(X-1)) = n(n-1)p^2$$

$$V(X) = np(1-p)$$

• with replacement

Additional:
disjoint if $A_1 \cap A_2 = \emptyset$
exhaustive if $A_1 \cup A_2 = \Omega$

⑤ Poisson RV

$$N \triangleq P_n(\lambda)$$

num of events per unit of time

$$P_N(k) = \frac{e^{-\lambda} \lambda^k}{k!}, k=0,1,2,\dots$$

$$E(X) = \lambda$$

$$E(X(X-1)) = \lambda^2$$

$$V(X) = \lambda$$

• with replacement

• num of events occurred by time 1

Binomial Approximation

$$\text{Bin}(n, p) \approx P_n(np)$$

for small $p \leq 0.05$

⑨ Gamma RV

$$Z \triangleq \gamma(r, \alpha)$$

$$F_Z(z) = P(Z \leq z)$$

$$= \begin{cases} 1 - \sum_{k=0}^{r-1} \frac{(\alpha z)^k}{k!} e^{-\alpha z}, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$f_Z(z) = \frac{\alpha^r z^{r-1}}{(r-1)!} e^{-\alpha z}$$

$$* (r-1)! \text{ is } \Gamma(r) \\ = (r-1)! \Gamma(r-1) \text{ for } r > 0$$

$$E(X^k) = \frac{\Gamma(r+k)}{\Gamma(r) \alpha^k}$$

$$E(X) = \frac{r}{\alpha}$$

$$\text{Var}(X) = \frac{r}{\alpha^2}$$

• continuous 版 negative binomial RV

• models waiting time until rth occurrence of an event in continuous time

例: 5 matrix with 5列 - 5行

`matrix(c(0), nrow=10000, ncol=2)`

`rnorm(n, mean=1, sd=sqrt(2))`

`m[,1] = rnorm(10000, 1, sqrt(2))`

用 rnorm 填满第一列 - 5行

② Negative binomial RV

$$X \triangleq \text{Nb}(r, p)$$

$$P_X(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$$

$x=0,1,2,\dots$

$$E(X) = \frac{r(1-p)}{p}$$

$$V(X) = \frac{r(1-p)}{p^2}$$

• with replacement

• num of failures before rth success

$$P_X(x) = \binom{x+r-1}{r-1} p^r (1-p)^x$$

⑥ Discrete uniform RV

$$X \triangleq U(m, n)$$

$$P_X(x) = \frac{1}{n-m+1}$$

$$E(X) = \frac{n}{2}$$

$$E(X^2) = \frac{1}{6} n(n+1)$$

$$V(X) = \frac{1}{12} n(n+1)$$

• with replacement

Conditional expectation
 $E(g(X)|Y=y) = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x|y) dx$

⑩ Beta RV

$$X \triangleq \text{Beta}(\alpha, \beta)$$

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)}, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

where

$$B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$E(X) = \frac{\alpha}{\alpha+\beta}$$

$$E(X^k) = \frac{\Gamma(\alpha+k) \Gamma(\alpha+\beta)}{\Gamma(\alpha) \Gamma(\alpha+\beta+k)}$$

$$V(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

• 被两个值 bounded
如 the voted population in Aus

⑬ Weibull RV

$$X \triangleq \text{Weibull}(\beta, \eta)$$

$$f_X(x) = \frac{\eta x^{\eta-1}}{\beta^\eta} e^{-(x/\beta)^\eta}, 0 \leq x < \infty$$

$$E(X) = \beta \Gamma((r+1)/\eta)$$

$$E(X^2) = \beta^2 \Gamma((r+2)/\eta)$$

$$\text{Var}(X) = \beta^2 \left(\Gamma((r+2)/\eta) - (\Gamma((r+1)/\eta))^2 \right)$$

③ Geometric RV

$$N \triangleq \text{Geom}(p), \lambda = \{0,1,2,\dots\}$$

$$P_N(n) = (1-p)^{n-1} p$$

$$E(X) = \frac{1}{p}$$

$$V(X) = \frac{1-p}{p^2}$$

$\frac{1}{p}$ [number of failure until first success]

• with replacement

• num of failures before first success

$$P(N \geq n) = (1-p)^{n-1}$$

$$P(N \geq n) = (1-p)^{\min(n, X_2)}$$

⑦ Continuous Uniform RV

$$X \triangleq U(a, b)$$

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$$

$$E(X) = \frac{a+b}{2}$$

$$\text{Var}(X) = \frac{1}{12} (b-a)^2$$

• sample once

⑬ Bivariate standard normal

$$f_{X|Y}(x|y) = \frac{1}{\sqrt{2\pi} \sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x-\rho y)^2}$$

⑪ Pareto RV

$$X \triangleq \text{Pareto}(\alpha, \eta)$$

$$f_X(x) = \frac{\eta \alpha^\eta}{x^{\eta+1}}$$

$$F_X(x) = \begin{cases} 1 - (\frac{\alpha}{x})^\eta, & x \geq \alpha \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \frac{\eta \alpha}{\eta-1}, \eta > 1$$

$$E(X^2) = \frac{\alpha^2 \eta}{\eta-2}, \eta > 2$$

$$V(X) = \frac{\eta \alpha^2}{(\eta-1)^2(\eta-2)}, \eta > 2$$

⑭ Bivariate Conditional

$$P(X \geq 3 | Y=y) = \int_3^\infty f_{X|Y}(x|y) dx$$

$$f_{X|Y}(x|y) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{(x^2-2\rho xy+y^2)}{2(1-\rho^2)}}$$

Format error

$$F_X(x) = \begin{cases} 0, & x < 0 \\ -, & 0 \leq x < 100 \\ 1, & x \geq 100 \end{cases}$$

$$f_X(x) = \begin{cases} -, & 0 \leq x < 100 \\ 0, & \text{otherwise} \end{cases}$$

Rstudio

Sample (data, size, replace=TRUE, prob=prob)

例 result = matrix(sample(c(0,1), 1000, replace=TRUE, prob=c(1/30, 1/30)))

生成 1000 行 0 或 1 的 value

④ Hypergeometric

$$N \triangleq \text{Hg}(n, d, \lambda)$$

sample size, total defects, total items

$$P_X(x) = \frac{\binom{d}{x} \binom{n-d}{n-x}}{\binom{n}{n}}$$

$$E(X) = \frac{nd}{n}$$

$$V(X) = \frac{nd(n-d)}{n^2} x(1-\frac{n-1}{n-1})$$

• without replacement

• num of defectives in sample of n

⑧ Exponential RV

$$T \triangleq \text{exp}(\alpha)$$

$$F_T(t) = 1 - e^{-\alpha t}, t \geq 0$$

$$f_T(t) = \begin{cases} \alpha e^{-\alpha t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$E(X) = \frac{1}{\alpha}$$

$$E(X^2) = \frac{2}{\alpha^2}$$

$$\text{Var}(X) = \frac{1}{\alpha^2}$$

• continuous 版 geometric RV
wait time until an event occur

⑫ Normal Distribution

$$* \therefore X \triangleq N(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$x \in \mathbb{R}$

Standardise

$$Z = \frac{X-\mu}{\sigma} \triangleq N(0, 1)$$

$$\text{例: } P(X < 8) = P(Z < 0.4)$$

$$* \therefore Z \triangleq N(0, 1)$$

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, z \in \mathbb{R}$$

$$E(Z) = \int_{-\infty}^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$E(Z^2) = \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$E(Z^{2k}) = \frac{2^k k!}{\sqrt{2\pi}}$$

Normal Approximation

$$X \triangleq \text{Bi}(n, p), n \text{ is large}$$

$$p \text{ 不近 } 0 \text{ 或 } 1$$

$$\text{例 } X \triangleq N(np, np(1-p))$$

$$X \triangleq P_n(\lambda), \lambda \text{ is large}$$

$$\text{例 } X \triangleq N(\lambda, \lambda)$$

$$X \triangleq \gamma(r, \alpha), r \text{ is large}$$

$$\text{例 } X \triangleq N(\frac{r}{\alpha}, \frac{r}{\alpha^2})$$

Axioms

- $P(A) \geq 0$, for all events A
- $P(\Omega) = 1$
- For disjoint events

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Some properties

$$A \subseteq B, \text{ then } B = A \cup (B \setminus A)$$

$$5. B \cap A = \emptyset \Rightarrow A = (A \cap B) \cup (A \setminus B)$$

$$\int_{-\infty}^{\infty} e^{-\frac{1}{2}x^2} dx = \sqrt{2\pi}$$

$$\Rightarrow Y \stackrel{d}{=} LN(\mu, \sigma^2)$$

$$E(Y) = e^{\mu + \frac{1}{2}\sigma^2}$$

$$E(Y^2) = e^{2\mu + 2\sigma^2}$$

$$Var(Y) = e^{2\mu + 2\sigma^2} (e^{\sigma^2} - 1)$$

$$E(Y^3) = e^{3\mu + \frac{3}{2}\sigma^2} (1 + 3\sigma^2)$$

$$E(Y^4) = e^{4\mu + 2\sigma^2} (1 + 6\sigma^2 + 3\sigma^4)$$

$$E(Y^5) = e^{5\mu + \frac{5}{2}\sigma^2} (1 + 10\sigma^2 + 10\sigma^4 + \frac{5}{2}\sigma^6)$$

$$E(Y^6) = e^{6\mu + 3\sigma^2} (1 + 15\sigma^2 + 45\sigma^4 + 15\sigma^6 + \frac{15}{2}\sigma^8)$$

$$E(Y^7) = e^{7\mu + \frac{7}{2}\sigma^2} (1 + 21\sigma^2 + 105\sigma^4 + 35\sigma^6 + \frac{35}{2}\sigma^8 + \frac{7}{2}\sigma^{10})$$

$$E(Y^8) = e^{8\mu + 4\sigma^2} (1 + 28\sigma^2 + 378\sigma^4 + 252\sigma^6 + 70\sigma^8 + \frac{35}{2}\sigma^{10} + \frac{7}{8}\sigma^{12})$$

$$E(Y^9) = e^{9\mu + \frac{9}{2}\sigma^2} (1 + 36\sigma^2 + 1260\sigma^4 + 1260\sigma^6 + 420\sigma^8 + \frac{315}{2}\sigma^{10} + \frac{315}{8}\sigma^{12} + \frac{63}{8}\sigma^{14})$$

$$E(Y^{10}) = e^{10\mu + 5\sigma^2} (1 + 45\sigma^2 + 4725\sigma^4 + 6720\sigma^6 + 20790\sigma^8 + 27720\sigma^{10} + 9240\sigma^{12} + \frac{10395}{2}\sigma^{14} + \frac{10395}{8}\sigma^{16} + \frac{10395}{128}\sigma^{18})$$

$$E(Y^{11}) = e^{11\mu + \frac{11}{2}\sigma^2} (1 + 55\sigma^2 + 10395\sigma^4 + 20790\sigma^6 + 103950\sigma^8 + 277200\sigma^{10} + 527040\sigma^{12} + 792000\sigma^{14} + \frac{103950}{2}\sigma^{16} + \frac{103950}{8}\sigma^{18} + \frac{103950}{128}\sigma^{20})$$

$$E(Y^{12}) = e^{12\mu + 6\sigma^2} (1 + 66\sigma^2 + 20790\sigma^4 + 67200\sigma^6 + 1039500\sigma^8 + 4200000\sigma^{10} + 10395000\sigma^{12} + 20790000\sigma^{14} + 27720000\sigma^{16} + 20790000\sigma^{18} + \frac{10395000}{2}\sigma^{20} + \frac{10395000}{8}\sigma^{22} + \frac{10395000}{128}\sigma^{24})$$

$$E(Y^{13}) = e^{13\mu + \frac{13}{2}\sigma^2} (1 + 78\sigma^2 + 47250\sigma^4 + 207900\sigma^6 + 10395000\sigma^8 + 52704000\sigma^{10} + 103950000\sigma^{12} + 207900000\sigma^{14} + 277200000\sigma^{16} + 207900000\sigma^{18} + \frac{103950000}{2}\sigma^{20} + \frac{103950000}{8}\sigma^{22} + \frac{103950000}{128}\sigma^{24} + \frac{103950000}{2048}\sigma^{26})$$

$$E(Y^{14}) = e^{14\mu + 7\sigma^2} (1 + 91\sigma^2 + 103950\sigma^4 + 672000\sigma^6 + 10395000\sigma^8 + 52704000\sigma^{10} + 103950000\sigma^{12} + 207900000\sigma^{14} + 277200000\sigma^{16} + 207900000\sigma^{18} + \frac{103950000}{2}\sigma^{20} + \frac{103950000}{8}\sigma^{22} + \frac{103950000}{128}\sigma^{24} + \frac{103950000}{2048}\sigma^{26} + \frac{103950000}{32768}\sigma^{28})$$

$$E(Y^{15}) = e^{15\mu + \frac{15}{2}\sigma^2} (1 + 105\sigma^2 + 207900\sigma^4 + 1039500\sigma^6 + 10395000\sigma^8 + 52704000\sigma^{10} + 103950000\sigma^{12} + 207900000\sigma^{14} + 277200000\sigma^{16} + 207900000\sigma^{18} + \frac{103950000}{2}\sigma^{20} + \frac{103950000}{8}\sigma^{22} + \frac{103950000}{128}\sigma^{24} + \frac{103950000}{2048}\sigma^{26} + \frac{103950000}{32768}\sigma^{28} + \frac{103950000}{524288}\sigma^{30})$$

$$E(Y^{16}) = e^{16\mu + 8\sigma^2} (1 + 120\sigma^2 + 1039500\sigma^4 + 6720000\sigma^6 + 103950000\sigma^8 + 527040000\sigma^{10} + 1039500000\sigma^{12} + 2079000000\sigma^{14} + 2772000000\sigma^{16} + 2079000000\sigma^{18} + \frac{1039500000}{2}\sigma^{20} + \frac{1039500000}{8}\sigma^{22} + \frac{1039500000}{128}\sigma^{24} + \frac{1039500000}{2048}\sigma^{26} + \frac{1039500000}{32768}\sigma^{28} + \frac{1039500000}{524288}\sigma^{30} + \frac{1039500000}{8388608}\sigma^{32})$$

$$E(Y^{17}) = e^{17\mu + \frac{17}{2}\sigma^2} (1 + 136\sigma^2 + 2079000\sigma^4 + 10395000\sigma^6 + 103950000\sigma^8 + 527040000\sigma^{10} + 1039500000\sigma^{12} + 2079000000\sigma^{14} + 2772000000\sigma^{16} + 2079000000\sigma^{18} + \frac{1039500000}{2}\sigma^{20} + \frac{1039500000}{8}\sigma^{22} + \frac{1039500000}{128}\sigma^{24} + \frac{1039500000}{2048}\sigma^{26} + \frac{1039500000}{32768}\sigma^{28} + \frac{1039500000}{524288}\sigma^{30} + \frac{1039500000}{8388608}\sigma^{32} + \frac{1039500000}{134217728}\sigma^{34})$$

$$E(Y^{18}) = e^{18\mu + 9\sigma^2} (1 + 153\sigma^2 + 20790000\sigma^4 + 67200000\sigma^6 + 1039500000\sigma^8 + 5270400000\sigma^{10} + 10395000000\sigma^{12} + 20790000000\sigma^{14} + 27720000000\sigma^{16} + 20790000000\sigma^{18} + \frac{10395000000}{2}\sigma^{20} + \frac{10395000000}{8}\sigma^{22} + \frac{10395000000}{128}\sigma^{24} + \frac{10395000000}{2048}\sigma^{26} + \frac{10395000000}{32768}\sigma^{28} + \frac{10395000000}{524288}\sigma^{30} + \frac{10395000000}{8388608}\sigma^{32} + \frac{10395000000}{134217728}\sigma^{34} + \frac{10395000000}{2147483648}\sigma^{36})$$

$$E(Y^{19}) = e^{19\mu + \frac{19}{2}\sigma^2} (1 + 171\sigma^2 + 207900000\sigma^4 + 1039500000\sigma^6 + 10395000000\sigma^8 + 52704000000\sigma^{10} + 103950000000\sigma^{12} + 207900000000\sigma^{14} + 277200000000\sigma^{16} + 207900000000\sigma^{18} + \frac{103950000000}{2}\sigma^{20} + \frac{103950000000}{8}\sigma^{22} + \frac{103950000000}{128}\sigma^{24} + \frac{103950000000}{2048}\sigma^{26} + \frac{103950000000}{32768}\sigma^{28} + \frac{103950000000}{524288}\sigma^{30} + \frac{103950000000}{8388608}\sigma^{32} + \frac{103950000000}{134217728}\sigma^{34} + \frac{103950000000}{2147483648}\sigma^{36} + \frac{103950000000}{34359738368}\sigma^{38})$$

$$E(Y^{20}) = e^{20\mu + 10\sigma^2} (1 + 190\sigma^2 + 2079000000\sigma^4 + 10395000000\sigma^6 + 103950000000\sigma^8 + 527040000000\sigma^{10} + 1039500000000\sigma^{12} + 2079000000000\sigma^{14} + 2772000000000\sigma^{16} + 2079000000000\sigma^{18} + \frac{1039500000000}{2}\sigma^{20} + \frac{1039500000000}{8}\sigma^{22} + \frac{1039500000000}{128}\sigma^{24} + \frac{1039500000000}{2048}\sigma^{26} + \frac{1039500000000}{32768}\sigma^{28} + \frac{1039500000000}{524288}\sigma^{30} + \frac{1039500000000}{8388608}\sigma^{32} + \frac{1039500000000}{134217728}\sigma^{34} + \frac{1039500000000}{2147483648}\sigma^{36} + \frac{1039500000000}{34359738368}\sigma^{38} + \frac{1039500000000}{549753830400}\sigma^{40})$$

$$E(Y^{21}) = e^{21\mu + \frac{21}{2}\sigma^2} (1 + 210\sigma^2 + 20790000000\sigma^4 + 103950000000\sigma^6 + 1039500000000\sigma^8 + 5270400000000\sigma^{10} + 10395000000000\sigma^{12} + 20790000000000\sigma^{14} + 27720000000000\sigma^{16} + 20790000000000\sigma^{18} + \frac{10395000000000}{2}\sigma^{20} + \frac{10395000000000}{8}\sigma^{22} + \frac{10395000000000}{128}\sigma^{24} + \frac{10395000000000}{2048}\sigma^{26} + \frac{10395000000000}{32768}\sigma^{28} + \frac{10395000000000}{524288}\sigma^{30} + \frac{10395000000000}{8388608}\sigma^{32} + \frac{10395000000000}{134217728}\sigma^{34} + \frac{10395000000000}{2147483648}\sigma^{36} + \frac{10395000000000}{34359738368}\sigma^{38} + \frac{10395000000000}{549753830400}\sigma^{40} + \frac{10395000000000}{8589973760000}\sigma^{42})$$

$$E(Y^{22}) = e^{22\mu + 11\sigma^2} (1 + 231\sigma^2 + 207900000000\sigma^4 + 1039500000000\sigma^6 + 10395000000000\sigma^8 + 52704000000000\sigma^{10} + 103950000000000\sigma^{12} + 207900000000000\sigma^{14} + 277200000000000\sigma^{16} + 207900000000000\sigma^{18} + \frac{103950000000000}{2}\sigma^{20} + \frac{103950000000000}{8}\sigma^{22} + \frac{103950000000000}{128}\sigma^{24} + \frac{103950000000000}{2048}\sigma^{26} + \frac{103950000000000}{32768}\sigma^{28} + \frac{103950000000000}{524288}\sigma^{30} + \frac{103950000000000}{8388608}\sigma^{32} + \frac{103950000000000}{134217728}\sigma^{34} + \frac{103950000000000}{2147483648}\sigma^{36} + \frac{103950000000000}{34359738368}\sigma^{38} + \frac{103950000000000}{549753830400}\sigma^{40} + \frac{103950000000000}{8589973760000}\sigma^{42} + \frac{103950000000000}{134217728000000}\sigma^{44})$$

$$E(Y^{23}) = e^{23\mu + \frac{23}{2}\sigma^2} (1 + 253\sigma^2 + 2079000000000\sigma^4 + 10395000000000\sigma^6 + 103950000000000\sigma^8 + 527040000000000\sigma^{10} + 1039500000000000\sigma^{12} + 2079000000000000\sigma^{14} + 2772000000000000\sigma^{16} + 2079000000000000\sigma^{18} + \frac{1039500000000000}{2}\sigma^{20} + \frac{1039500000000000}{8}\sigma^{22} + \frac{1039500000000000}{128}\sigma^{24} + \frac{1039500000000000}{2048}\sigma^{26} + \frac{1039500000000000}{32768}\sigma^{28} + \frac{1039500000000000}{524288}\sigma^{30} + \frac{1039500000000000}{8388608}\sigma^{32} + \frac{1039500000000000}{134217728}\sigma^{34} + \frac{1039500000000000}{2147483648}\sigma^{36} + \frac{1039500000000000}{34359738368}\sigma^{38} + \frac{1039500000000000}{549753830400}\sigma^{40} + \frac{1039500000000000}{8589973760000}\sigma^{42} + \frac{1039500000000000}{134217728000000}\sigma^{44} + \frac{1039500000000000}{2147483648000000}\sigma^{46})$$

$$E(Y^{24}) = e^{24\mu + 12\sigma^2} (1 + 276\sigma^2 + 20790000000000\sigma^4 + 103950000000000\sigma^6 + 1039500000000000\sigma^8 + 5270400000000000\sigma^{10} + 10395000000000000\sigma^{12} + 20790000000000000\sigma^{14} + 27720000000000000\sigma^{16} + 20790000000000000\sigma^{18} + \frac{10395000000000000}{2}\sigma^{20} + \frac{10395000000000000}{8}\sigma^{22} + \frac{10395000000000000}{128}\sigma^{24} + \frac{10395000000000000}{2048}\sigma^{26} + \frac{10395000000000000}{32768}\sigma^{28} + \frac{10395000000000000}{524288}\sigma^{30} + \frac{10395000000000000}{8388608}\sigma^{32} + \frac{10395000000000000}{134217728}\sigma^{34} + \frac{10395000000000000}{2147483648}\sigma^{36} + \frac{10395000000000000}{34359738368}\sigma^{38} + \frac{10395000000000000}{549753830400}\sigma^{40} + \frac{10395000000000000}{8589973760000}\sigma^{42} + \frac{10395000000000000}{134217728000000}\sigma^{44} + \frac{10395000000000000}{2147483648000000}\sigma^{46} + \frac{10395000000000000}{34359738368000000}\sigma^{48})$$

$$E(Y^{25}) = e^{25\mu + \frac{25}{2}\sigma^2} (1 + 300\sigma^2 + 207900000000000\sigma^4 + 1039500000000000\sigma^6 + 10395000000000000\sigma^8 + 52704000000000000\sigma^{10} + 103950000000000000\sigma^{12} + 207900000000000000\sigma^{14} + 277200000000000000\sigma^{16} + 207900000000000000\sigma^{18} + \frac{103950000000000000}{2}\sigma^{20} + \frac{103950000000000000}{8}\sigma^{22} + \frac{103950000000000000}{128}\sigma^{24} + \frac{103950000000000000}{2048}\sigma^{26} + \frac{103950000000000000}{32768}\sigma^{28} + \frac{103950000000000000}{524288}\sigma^{30} + \frac{103950000000000000}{8388608}\sigma^{32} + \frac{103950000000000000}{134217728}\sigma^{34} + \frac{103950000000000000}{2147483648}\sigma^{36} + \frac{103950000000000000}{34359738368}\sigma^{38} + \frac{103950000000000000}{549753830400}\sigma^{40} + \frac{103950000000000000}{8589973760000}\sigma^{42} + \frac{103950000000000000}{134217728000000}\sigma^{44} + \frac{103950000000000000}{2147483648000000}\sigma^{46} + \frac{103950000000000000}{34359738368000000}\sigma^{48} + \frac{103950000000000000}{549753830400000000}\sigma^{50})$$

$$E(Y^{26}) = e^{26\mu + 13\sigma^2} (1 + 326\sigma^2 + 2079000000000000\sigma^4 + 10395000000000000\sigma^6 + 103950000000000000\sigma^8 + 527040000000000000\sigma^{10} + 1039500000000000000\sigma^{12} + 2079000000000000000\sigma^{14} + 2772000000000000000\sigma^{16} + 2079000000000000000\sigma^{18} + \frac{1039500000000000000}{2}\sigma^{20} + \frac{1039500000000000000}{8}\sigma^{22} + \frac{1039500000000000000}{128}\sigma^{24} + \frac{1039500000000000000}{2048}\sigma^{26} + \frac{1039500000000000000}{32768}\sigma^{28} + \frac{1039500000000000000}{524288}\sigma^{30} + \frac{1039500000000000000}{8388608}\sigma^{32} + \frac{1039500000000000000}{134217728}\sigma^{34} + \frac{1039500000000000000}{2147483648}\sigma^{36} + \frac{1039500000000000000}{34359738368}\sigma^{38} + \frac{1039500000000000000}{549753830400}\sigma^{40} + \frac{1039500000000000000}{8589973760000}\sigma^{42} + \frac{1039500000000000000}{134217728000000}\sigma^{44} + \frac{1039500000000000000}{2147483648000000}\sigma^{46} + \frac{1039500000000000000}{34359738368000000}\sigma^{48} + \frac{1039500000000000000}{549753830400000000}\sigma^{50} + \frac{1039500000000000000}{858997376000000000}\sigma^{52})$$

$$E(Y^{27}) = e^{27\mu + \frac{27}{2}\sigma^2} (1 + 354\sigma^2 + 20790000000000000\sigma^4 + 103950000000000000\sigma^6 + 1039500000000000000\sigma^8 + 5270400000000000000\sigma^{10} + 10395000000000000000\sigma^{12} + 20790000000000000000\sigma^{14} + 27720000000000000000\sigma^{16} + 20790000000000000000\sigma^{18} + \frac{10395000000000000000}{2}\sigma^{20} + \frac{10395000000000000000}{8}\sigma^{22} + \frac{10395000000000000000}{128}\sigma^{24} + \frac{10395000000000000000}{2048}\sigma^{26} + \frac{10395000000000000000}{32768}\sigma^{28} + \frac{10395000000000000000}{524288}\sigma^{30} + \frac{10395000000000000000}{8388608}\sigma^{32} + \frac{10395000000000000000}{134217728}\sigma^{34} + \frac{10395000000000000000}{2147483648}\sigma^{36} + \frac{10395000000000000000}{34359738368}\sigma^{38} + \frac{10395000000000000000}{549753830400}\sigma^{40} + \frac{10395000000000000000}{8589973760000}\sigma^{42} + \frac{10395000000000000000}{134217728000000}\sigma^{44} + \frac{10395000000000000000}{2147483648000000}\sigma^{46} + \frac{10395000000000000000}{34359738368000000}\sigma^{48} + \frac{10395000000000000000}{549753830400000000}\sigma^{50} + \frac{10395000000000000000}{858997376000000000}\sigma^{52} + \frac{10395000000000000000}{1342177280000000000}\sigma^{54})$$

$$E(Y^{28}) = e^{28\mu + 14\sigma^2} (1 + 384\sigma^2 + 207900000000000000\sigma^4 + 1039500000000000000\sigma^6 + 10395000000000000000\sigma^8 + 52704000000000000000\sigma^{10} + 103950000000000000000\sigma^{12} + 207900000000000000000\sigma^{14} + 277200000000000000000\sigma^{16} + 207900000000000000000\sigma^{18} + \frac{103950000000000000000}{2}\sigma^{20} + \frac{103950000000000000000}{8}\sigma^{22} + \frac{103950000000000000000}{128}\sigma^{24} + \frac{103950000000000000000}{2048}\sigma^{26} + \frac{103950000000000000000}{32768}\sigma^{28} + \frac{103950000000000000000}{524288}\sigma^{30} + \frac{103950000000000000000}{8388608}\sigma^{32} + \frac{103950000000000000000}{134217728}\sigma^{34} + \frac{103950000000000000000}{2147483648}\sigma^{36} + \frac{103950000000000000000}{34359738368}\sigma^{38} + \frac{103950000000000000000}{549753830400}\sigma^{40} + \frac{103950000000000000000}{8589973760000}\sigma^{42} + \frac{103950000000000000000}{134217728000000}\sigma^{44} + \frac{103950000000000000000}{2147483648000000}\sigma^{46} + \frac{103950000000000000000}{34359738368000000}\sigma^{48} + \frac{103950000000000000000}{549753830400000000}\sigma^{50} + \frac{103950000000000000000}{858997376000000000}\sigma^{52} + \frac{103950000000000000000}{1342177280000000000}\sigma^{54} + \frac{103950000000000000000}{21474836480000000000}\sigma^{56})$$

$$E(Y^{29}) = e^{29\mu + \frac{29}{2}\sigma^2} (1 + 416\sigma^2 + 2079000000000000000\sigma^4 + 10395000000000000000\sigma^6 + 103950000000000000000\sigma^8 + 5270400000$$